

T3

Discretization & Directional Quantization

Lion

S1 scope: element-wise parameters

S2 signal: sign of momentum-gradient interpolation

S3 state: one momentum buffer

S4 reconstruct: identity reconstruction

S5 update: signed step + decoupled weight decay

Ano

+ Restore magnitude
S2/S5: sign direction +
magnitude channel

FOCUS

+ Add attraction
S5: moving-average
attraction term

RLion

+ Smooth design
S2: smooth sign-like
map

SignSGD

+ Add momentum
S2: hard sign of raw gradient

Signum

+ Modern interface
S2/S3: sign of
momentum buffer

